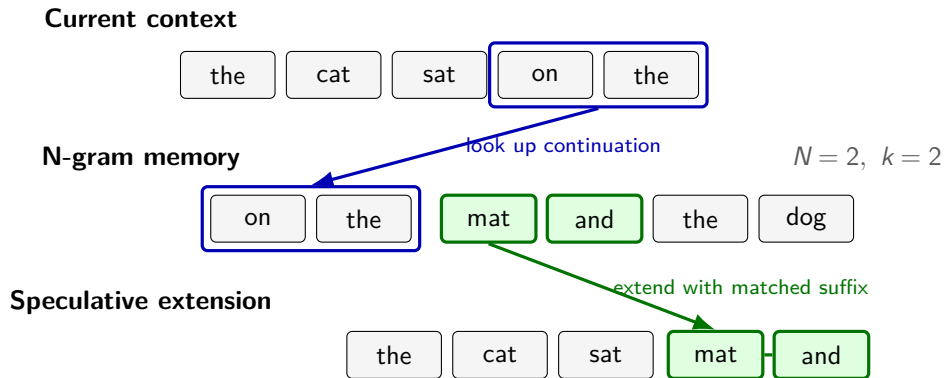


N-gram Speculative Decoding Implementation

Jia Guo

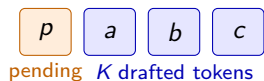
N-gram matching and extending



If the recent suffix appears in memory, reuse its following tokens as a draft; the target model verifies them in one pass.

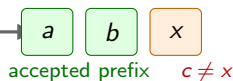
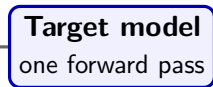
Speculative decoding: greedy verification and KV cleanup

1. N-gram draft



Verification input:
 $p + [a, b, c]$

2. One forward 3. Greedy commit



$K + 1$ greedy predictions Replace the first mismatched draft
with the target-model token.

KV-cache update

Keep cached: $[\dots, p, a, b]$



`page_size = 1` makes rejected-draft cleanup token-granular;
the final emitted token remains pending without KV cache.

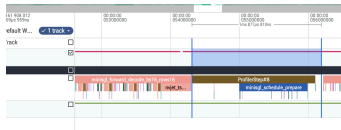
Design decisions

- ▶ **Request-local lookup:** search only the current request's context.
- ▶ **N-gram fallback:** if $N = 3$ has no match, retry $N = 2$, then $N = 1$.
- ▶ **Page size = 1:** Mini-SGLang's default; enables token-level KV-cache cleanup after rejected drafts.
- ▶ **Fold decode into verification:** scheduling has prefill and verification steps only—no separate decode step.

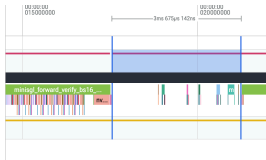
Verification and forward-prep optimizations

Four profiler snapshots across the optimization sequence.

Baseline — pure decoding



2. Faster n-gram lookup + batched draft-token $H \rightarrow D$



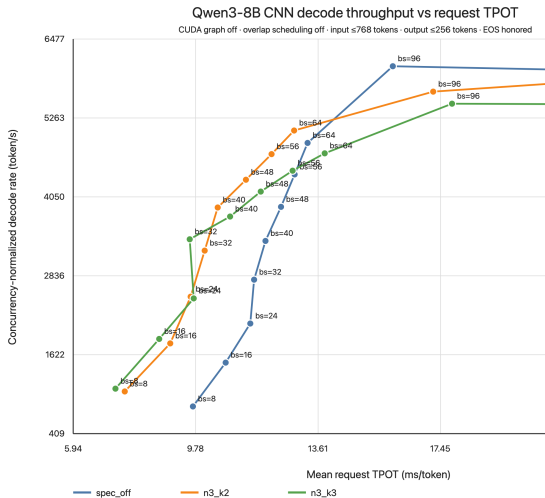
1. Vanilla speculative decoding



3. Batch all verification $H \rightarrow D$ + misc. optimizations



CNN summarization Decode throughput vs. request TPOT



CNN summarization Performance summary

Relative to pure decoding at the same request concurrency

Concurrency	N=3, K=2		N=3, K=3	
	Throughput	TPOT	Throughput	TPOT
8	+27.8%	+22.0%	+33.0%	+25.1%
16	+19.7%	+16.2%	+24.3%	+19.4%
24	+19.6%	+16.3%	+18.4%	+15.4%
32	+16.1%	+13.3%	+22.4%	+17.4%
40	+15.2%	+12.5%	+11.1%	+9.3%
48	+10.7%	+8.8%	+6.0%	+5.1%
56	+7.1%	+5.6%	+1.3%	+0.5%
64	+3.9%	+3.2%	-3.3%	-4.0%
96	-6.4%	-7.9%	-9.5%	-11.6%
128	-2.7%	-3.1%	-8.8%	-10.0%

Acceptance statistics

Draft length	Mean accepted	p0	p1	p2
$K = 2$	0.52	32.78%	57.27%	—
$K = 3$	0.59	31.22%	57.27%	57.93%

Mean accepted = draft tokens per verification step. TPOT = mean request time per output token. Later draft-position acceptance is conditional on every earlier position being accepted.

Numerical test results

Test configuration

- ▶ FlashInfer backend
- ▶ 2,000 CNN/DailyMail cases
- ▶ Batch size 8; greedy decoding
- ▶ $N = 3$, $K = 4$; 32 generated tokens
- ▶ EOS ignored

Observed against pure decoding

385 / 2,000 requests

had a token mismatch (**19.25%**); the strict exact-token test failed.

Compared token log-probabilities	59,066
Mean $ \Delta \log p $	0.00859
P99 $ \Delta \log p $	0.07795
Maximum $ \Delta \log p $	0.19228

Separate EOS-honored CNN run, batch size 128 (128 traces): exact completion matches were **84 / 128** for $N = 3$, $K = 2$ and **90 / 128** for $N = 3$, $K = 3$.

Batch-size sensitivity

Main SGLang reproduces the instability on **50 / 50 adversarial prompts** at batch size 1; batch size 8 is not stable. The effect is batch-size sensitive.

CNN exact mismatch example 1 (BS=128)

Article ID: 7fdbcd9bd3ea48687457817b6440183837d97c85. First differing character: 37.

Spec-off

- ▶ Daryl Janmaat compared Newcastle's upcoming Premier League match against Swansea to a World Cup semi-final due to its significance.
- ▶ Newcastle has lost their last six league games, leaving them seven points above the relegation zone with five games remaining.
- ▶ Janmaat, who played in the 2014 World Cup semi-final, emphasized the importance of winning to ensure safety from relegation.

$N = 3$, $K = 2$ speculative

- ▶ Daryl Janmaat compared Newcastle's Premier League match against Swansea to a World Cup semi-final due to its significance.
- ▶ Newcastle has lost their last six league games and is seven points above the relegation zone with five games remaining.
- ▶ Janmaat, who played in the 2014 World Cup semi-final, emphasized the importance of winning to ensure safety from relegation.

CNN exact mismatch: second example (BS=128)

Article ID: 0ecc8cb598f4bdee432dd9d5de63bd64d74eb794. First differing character: 45.

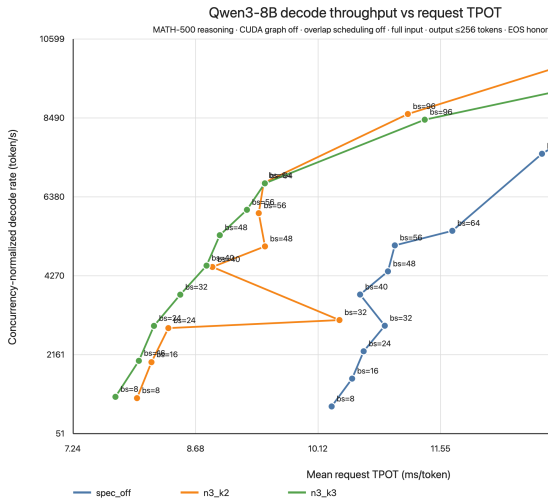
Spec-off

- ▶ James Lockwood received a two-year ban for doping after testing positive for Growth Hormone Releasing Peptide-2 (pralmorelin) in an out-of-competition test.
- ▶ The ban, the first of its kind in the UK, starts on 3 March 2015 and ends on 2 March 2017.
- ▶ Lockwood, a 29-year-old Featherstone player, joined the team in 2012 and was part of the squad that lost the Championship Grand Final in October 2014.

$N = 3$, $K = 2$ speculative

- ▶ James Lockwood received a two-year ban for failing a doping test, marking the first such case in the UK under the Rugby Football League's anti-doping regulations.
- ▶ The positive test result was for Growth Hormone Releasing Peptide-2 (pralmorelin) and its metabolite, identified in an out-of-competition sample from November 2014.
- ▶ Lockwood, a 29-year-old Featherstone prop, is suspended from all competition from 3 March 2015 to 2 March 2017.

MATH-500 decode throughput vs. request TPOT



MATH-500 reasoning performance summary

500 fixed problems per point; reasoning enabled; full input; EOS honored; CUDA graph and overlap scheduling disabled.

Relative to pure decoding at the same request concurrency

Concurrency	N=3, K=2		N=3, K=3	
	Throughput	TPOT	Throughput	TPOT
8	+28.6%	+22.2%	+32.8%	+24.7%
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32	+5.1%	+4.9%	+28.3%	+22.0%
40	+19.5%	+16.3%	+20.5%	+17.0%
48	+15.2%	+13.2%	+22.0%	+18.1%
56	+17.0%	+14.5%	+18.7%	+15.8%
64	+23.4%	+18.9%	+23.2%	+18.8%
96	+14.1%	+12.4%	+12.1%	+10.8%
128	+7.5%	+7.0%	+3.2%	+3.1%

Acceptance statistics

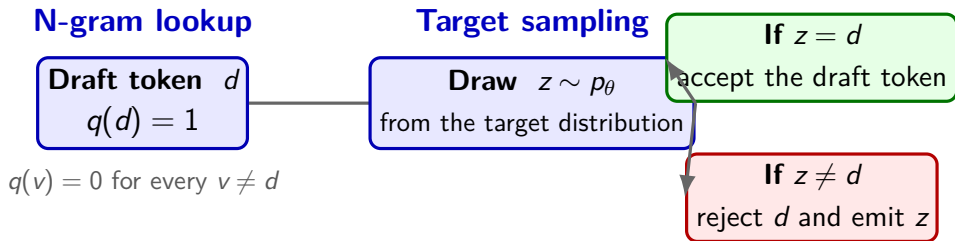
Draft length	Mean accepted	p0	p1	p2
$K = 2$	0.43	28.64%	52.05%	—
$K = 3$	0.49	27.23%	50.92%	59.17%

CUDA graphs + overlapped scheduling

CUDA graph enabled · overlap scheduling enabled



N-gram rejection sampling: deterministic proposal



Why $q = 1$ matters: $\Pr[\text{accept}] = p_\theta(d | h), \quad z | z \neq d \sim \frac{p_\theta(z | h)}{1 - p_\theta(d | h)} \quad (z \neq d)$

Apply the same rule left-to-right over the draft; stop at the first rejection.